

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
## $n.burnin
## [1] 1000
##
## $n.draw
## [1] 1200
##
## $thin
## [1] 1
##
## $pp_cutoff
## [1] 0.5
##
## $prior_gamma
## [1] 0.1 0.1 0.1 0.1 0.1
##
## $K_values
## [1] "1 / Examined number of clusters: 2" "2 / Examined number of clusters: 3"
## [3] "3 / Examined number of clusters: 4" "4 / Examined number of clusters: 5"
## [5] "5 / Examined number of clusters: 6" "6 / Examined number of clusters: 7"
## [7] "7 / Examined number of clusters: 8" "8 / Examined number of clusters: 9"
## [9] "9 / Examined number of clusters: 10"
##
## $sdev
## [1] 0.05
##
## $beta.var.scale
## [1] 0.3
```

Summary Statistics

```
--- Summary for K = 1 ---
Z.ar (Location: fit[[1]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.2117  0.2433  0.2525  0.2526  0.2617  0.3050

beta.ar (Location: fit[[1]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0950  0.1425  0.1381  0.1750  0.9225
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Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[1]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09833 0.14583 0.14092 0.17833 0.76250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.102175 0.030833 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.077168 0.023333 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005833 0.015000 0.162607 0.118333 1.000000

--- Summary for K = 2 ---
Z.ar (Location: fit[[2]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.04333 0.06083 0.06750 0.06746 0.07333 0.09250

beta.ar (Location: fit[[2]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.05667 0.10500 0.11000 0.14583 0.91667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[2]]$gamma.ar):
  Modality 1 :

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	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00000	0.05833	0.10833	0.11281	0.15000	0.74417
Modality 2 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.000000	0.001667	0.055833	0.413726	1.000000	1.000000
Modality 3 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 4 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000000	0.0008333	0.0050000	0.2487155	0.3897917	1.0000000
Modality 5 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.1200	1.0000	0.6968	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar (Location: fit[[3]]\$Z.ar):

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.01000	0.02000	0.02333	0.02338	0.02667	0.04083

beta.ar (Location: fit[[3]]\$beta.ar):

Modality 1 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00000	0.03417	0.07417	0.08367	0.11583	0.88417
Modality 2 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 3 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 4 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 5 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1

gamma.ar (Location: fit[[3]]\$gamma.ar):

Modality 1 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.00000	0.03667	0.07833	0.08695	0.12167	0.64667
Modality 2 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.9717	1.0000	0.7948	1.0000	1.0000
Modality 3 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1	1	1	1	1	1
Modality 4 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.0000	0.0000	0.2246	0.1669	1.0000
Modality 5 :						
	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.0000	0.0500	1.0000	0.6741	1.0000	1.0000

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--- Summary for K = 4 ---
Z.ar (Location: fit[[4]]$Z.ar):
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.003333 0.013333 0.018333 0.018900 0.022500 0.077500

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beta.ar (Location: fit[[4]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.025000 0.06083 0.07170 0.09917 0.84667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

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gamma.ar (Location: fit[[4]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.02667 0.06500 0.07543 0.10500 0.65667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.5333  1.0000  0.7355  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.0000  0.2573  0.5575  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  1.0000  1.0000  0.8376  1.0000  1.0000

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--- Summary for K = 5 ---
Z.ar (Location: fit[[5]]$Z.ar):
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.002500 0.009167 0.011667 0.012276 0.015000 0.076667

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beta.ar (Location: fit[[5]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.02167 0.05500 0.06575 0.09167 0.83333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

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Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[5]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02333 0.06000 0.07003 0.09833 0.59583
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.009167 0.704167 0.572172 1.000000 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2587 0.6046 1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8756 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar (Location: fit[[6]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0008333 0.0050000 0.0075000 0.0075827 0.0091667 0.0225000

beta.ar (Location: fit[[6]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01167 0.03750 0.05302 0.07500 0.83083
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[6]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01333 0.04167 0.05679 0.08167 0.60833
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.4115 1.0000 1.0000
  Modality 3 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7683	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2199	0.2058	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7758	1.0000	0.7881	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar (Location: fit[[7]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001667	0.005833	0.007500	0.008543	0.010833	0.039167

beta.ar (Location: fit[[7]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.009167	0.031667	0.046279	0.066667	0.776667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[7]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0100	0.0350	0.0504	0.0750	0.5300

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.6258	0.5227	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4567	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1958	0.0000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1283	0.9208	0.6547	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar (Location: fit[[8]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0008333	0.0066667	0.0083333	0.0089800	0.0108333	0.0225000

beta.ar (Location: fit[[8]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00750 0.02708 0.04081 0.05750 0.73250
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1

gamma.ar (Location: fit[[8]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.008333 0.031667 0.045802 0.066667 0.531667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2989 0.8042 1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 1.0000 1.0000 0.9965 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1093 0.0000 1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.6950 0.5424 1.0000 1.0000

--- Summary for K = 9 ---
Z.ar (Location: fit[[9]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.001667 0.006667 0.008333 0.009176 0.010833 0.024167

beta.ar (Location: fit[[9]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005833 0.025833 0.041742 0.059792 0.691667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1

```

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gamma.ar (Location: fit[[9]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.006667 0.030000 0.046973 0.070000 0.517500
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2273  0.4125  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  1.0000  1.0000  0.9982  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.1012  0.0000  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.6150  0.5228  1.0000  1.0000

```